PROJECT 30 | REVIEW 3 (WEEK 16 – FINAL PRESENTATION)

Sports Club Membership & Tournament Management System

Final System Architecture, Full-Stack Integration, CRUD Operations & Viva Demonstration

Review 3 Objectives & Final Deliverables

Summary of complete system integration, testing, and deployment milestones achieved.

Full-Stack Application

Java/Python frontend connected securely to the normalized MySQL/PostgreSQL backend database.

Complete CRUD Suite

Interactive modules for member registration, team setup, training schedules, and fixture management.

Robust Error Handling

Transaction safety, input validation, exception catch blocks, and automated integrity constraints.

System Architecture & Tech Stack

Structured layering ensuring modularity, secure database connection pools, and clean separation of concerns.

Presentation Layer

GUI / CLI: Python Tkinter or Java Swing/FX interface

Forms: Dynamic input forms with real-time validation

Dashboards: Summary widgets for dues and counts

Reports: Exportable tabular text/PDF summaries

Application Logic

Controllers: Business rule validation handlers

JDBC / ODBC: Secure database connector driver

Transactions: ACID compliance with rollback triggers

Security: Parameterized SQL queries against injection

Database Backend

DBMS: MySQL / PostgreSQL / Oracle Engine

Schema: Normalized 3NF tables (13 relations)

Views: Pre-compiled aggregation reporting views

Indexing: B-Tree indices on frequently searched keys

Interactive Functional Modules

Comprehensive workflow support across membership administration, scheduling, and events.

Membership & Fee Collection

Register members, assign subscription plans, record fee payments, and flag pending financial dues automatically.

Facility Booking & Training Scheduling

Schedule team training sessions, allocate sports facilities, and prevent double-booking conflicts via SQL exclusion rules.

Fixture Generation & Result Entry

Generate tournament match fixtures, log validated scores, and compute real-time team standings and leaderboards.

System Testing & Exception Handling

Rigorous validation protocols ensuring data integrity and resilient application execution.

Constraint Violations

Caught SQL Integrity Exceptions gracefully when attempting duplicate bookings, expired memberships, or capacity overflows.

Input Sanitization

Frontend form validators reject empty fields, malformed dates, negative fees, and invalid mobile number formats prior to query execution.

Transaction Rollbacks

Multi-step payment and equipment issue logs wrapped in atomic transaction blocks with automatic rollback on runtime failure.

PROJECT 30 COMPLETION

System Ready for Final Evaluation & Viva

All requirements from Conceptual Design through 3NF Normalization to the fully functional Java/Python application and test suite have been successfully fulfilled.

3NF
Normalized Database

100%
CRUD & Reporting Tests Passed

Week 16
Final Milestone Delivery